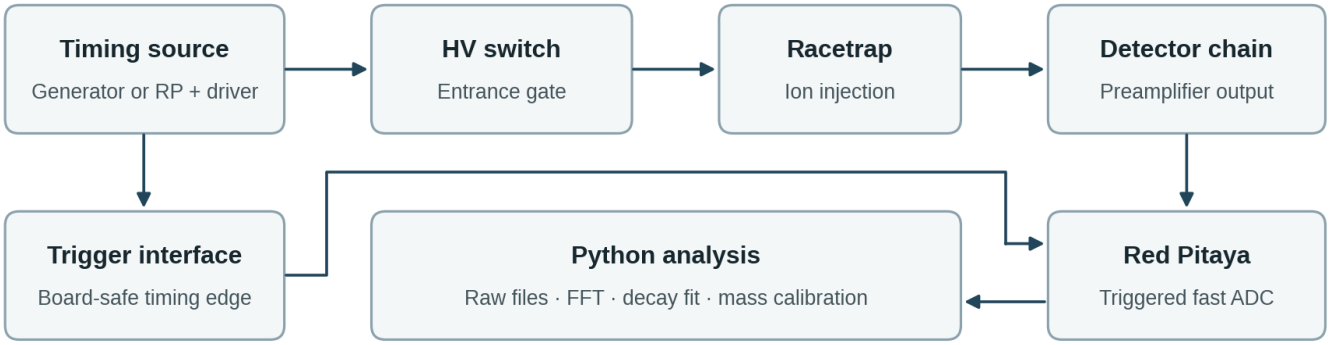
Racetrap Measurement Pipeline with Red Pitaya

Scope and implementation plan

Build a repeatable acquisition and analysis pipeline for the Racetrap V2 electrostatic ion trap. A hardware pulse initiates ion injection and gives the Red Pitaya a common timing reference. The script records the detector transient, extracts its frequency peaks and decay, and converts identified peaks to mass-to-charge ratio once an ion calibration is available.

Signal path



The timing and detector paths meet at the digitizer. With Red Pitaya as the pulse source, a supported internal generator trigger can provide the timing reference. RP means Red Pitaya. [1, 6]

Work to deliver

Acquisition script Arm the digitizer, coordinate the pulse, capture the requested interval, and save raw waveforms with the trigger index, hardware settings, and shot status.

Analysis and calibration Replay saved shots to produce a time trace, a DFT magnitude spectrum, a peak table, and a decay fit. Verify voltage and frequency readout with electrical references and use known ions to calibrate the mass axis.

Bench evidence Save one low-voltage validation run and one trap run with a matched background. Record the trigger timing, usable window, clipping checks, repeatability, and calibration residuals.

Project context

The September Racetrap V2 slides report improved signal-to-noise with differential detection, sodium signals beyond 20 ms, and an FFT example using a 10 ms Blackman–Harris window. The WhatsApp sketch proposes a roughly 100 ms timescale. The first build should reproduce the short acquisition before extending the capture. The longer-term pressure measurement goal needs its own gas calibration. [1–4]

Trigger and detector connections

For the first build, let the bench generator or pulse sequencer supply the timing edge. Send one branch to the existing switch control input and one branch through a board-compatible trigger interface to the Red Pitaya. Arm acquisition before enabling or requesting the pulse. [5, 6]

Signal in the sketch	Working interpretation and bench check
Timing source	Approximately 0–5 V and perhaps 3–5 μs wide. Confirm amplitude at the actual load, width, edge polarity and whether the nearby 100 ms label is the repetition interval.
Entrance electrodes	The sketch appears to show opposite biases around –200 V and +200 V moving toward 0 V for roughly 10 μs. Verify actual wiring, levels and pulse timing with the lab.
Detector transient	Record the output of the differential charge-sensitive preamplifier, with any required input conditioning. Measure switching pickup, recovery time, expected signal amplitude and noise.
Frequency and mass sketch	The labels appear to show potassium near nominal m/z 39 and about 120 kHz. Treat these as illustrative values pending an identified calibrant and a recorded trap configuration.

Electrical limits and timing

Identify the board model, hardware revision and installed Red Pitaya OS before wiring. For example, the original STEMlab 125-14 uses 3.3 V logic on E1 DIO0_P. Use a compatible level translator or trigger interface for the sketched 5 V signal. The documented acquisition debounce default is 500 μs. Verify detection and latency with the actual short pulse, then adjust the acquisition debounce or stretch only the digitizer branch if needed. [5, 6]

Keep the high-voltage electrode wiring on the switch side. Start with a low-voltage loopback and the high-voltage stage disabled. Confirm grounding, input range, coupling, load and worst-case overshoot before attaching the detector. For floating electrode measurements, follow the lab procedure and use a differential probe rated for both differential and common-mode voltages. [5, 9]

If Red Pitaya must issue the pulse, use a hardware-timed burst and an appropriate external driver for the switch input. Verify the loaded waveform on an oscilloscope. On supported configurations, use the internal generator event (AWG_PE) or a conditioned loopback for acquisition. Measure trigger-to-gate delay and jitter, and record the gate-close time when available. [6]

Sampling and acquisition script

Set the effective sample rate f_s to the board ADC clock divided by decimation D. A buffer of N samples covers approximately N/f_s seconds. Keep the highest frequency of interest below the Nyquist limit $f_s/2$ with room for the anti-alias filter. Verify the response of the analog front end and decimator to out-of-band signals, especially switch pickup. [5–7]

Capture mode	D	Sample rate	Nyquist limit	Record or samples per channel
Standard buffer	64	1.953 MS/s	0.977 MHz	16,384 samples cover 8.39 ms
Standard buffer	1024	122.1 kS/s	61.0 kHz	134.22 ms, excludes 120 kHz
Deep Memory	64	1.953 MS/s	0.977 MHz	At least 195,313 samples for 100 ms
Deep Memory	16	7.813 MS/s	3.906 MHz	781,250 samples for 100 ms

Illustrative values for a 125 MS/s model. Verify supported modes, buffer alignment and decimation on the actual board. Add samples for pretrigger, gate recovery and any other timing margin. [5–7]

Start with standard acquisition for a short electrical check. Use triggered Deep Memory Acquisition (DMA) for the longer run if supported. D = 64 is a starting point for peaks near 120 kHz. D = 16 leaves Nyquist margin if the experiment needs the broader band below 2 MHz described in the project notes. At 16-bit storage, the D = 16 example takes about 1.56 MB per channel per 100 ms shot. [3, 7]

One shot from arm to file

1. **Configure and arm** Set input range, sample rate, channel selection, trigger edge, and pre/post-trigger lengths. For the standard SCPI path, send ACQ:START, allow pretrigger fill, then send ACQ:TRig EXT_PE for an external rising edge. [6]

2. **Wait and read** Use separate timeouts for the trigger and buffer fill. The standard path can poll ACQ:TRig:STAT? and ACQ:TRig:FILL?. For DMA, check every enabled channel and use the documented buffer pointers to recover sample order and the trigger index. [6, 7]

3. **Save and rearm** Validate count, channel alignment and clipping. Save raw samples and metadata, then rearm. Log missed, partial and rejected shots. Use binary transfer and measure dead time. If the pulse period is 100 ms, a 100 ms post-trigger record leaves essentially no rearm margin. Use a longer or gated shot interval for the first run.

File contract Store raw ADC counts in NPZ with a JSON metadata file, or in HDF5. Record run and shot IDs, UTC time, trigger index, f_s, channel units and gain, ranges, board and OS, pulse and trap settings, pressure if available, and calibration ID. Export a peak CSV and plots separately.

Signal processing and decay

What the detector measures

A bunch of trapped ions induces charge on nearby electrodes as it passes the pickup. Repeated passages create a periodic signal. Differential pickup suppresses common-mode noise when the channels are well matched. A useful model for one isolated component is a damped sinusoid, where f is the detected frequency, τ is the amplitude decay time and b is the baseline. [2–4, 8]

$$x(t) = b + A_0 \exp(-t/\tau) \cos(2\pi f t + \phi)$$

Compute the frequency spectrum

1. **Prepare the transient** Keep the raw capture. Estimate and remove the DC baseline, preferably from clean pretrigger samples. Exclude the switch transient and preamplifier recovery using a measured blanking interval. Use the analog differential output or, if both detector channels are acquired, match gain and delay before subtraction.

2. **Apply the DFT** For M usable samples, compute the windowed DFT below with a real FFT. Start with a periodic Blackman–Harris window. This window family was used in the project and reduces leakage at the cost of broader peaks. In Python, use scipy.signal.windows.blackmanharris(M, sym=False), numpy.fft.rfft, and numpy.fft.rfftfreq. [4, 10]

$$X[k] = \Sigma\, w[n]\, x[n] \exp(-2\pi i k n/M) \quad \text{for } n = 0 \ldots M-1$$

$$f[k] = k\, f_s/M \qquad \Delta f = f_s/M \approx 1/T$$

3. **Report peaks** Use a measured background or local noise floor to set a peak threshold. Report frequency, magnitude or integrated area with its normalization, width, signal-to-noise ratio, and flags for pickup, harmonics or overlap. A 10 ms usable segment gives 100 Hz bin spacing. A 100 ms usable segment gives 10 Hz. Windowing and decay determine the practical peak width. Zero padding interpolates the spectrum on a finer frequency grid.

For a bin-centered steady tone, a one-sided peak-amplitude spectrum can use $2|X[k]|/\Sigma w[n]$, with no doubling at DC or at Nyquist when present. For decaying ion signals, use the same window, fit interval and amplitude or area convention for calibration and measurement.

Fit the decay

Estimate the envelope of a single isolated peak by band-pass filtering and taking the Hilbert magnitude, or by demodulating around its frequency. For an exponential amplitude, $\ln A(t) = \ln A_0 - t/\tau$. A fit slope s therefore gives $\tau = -1/s$ and decay rate $\gamma = -s$, in s^{-1}. If using power instead of amplitude, the log-power slope is $-2/\tau$.

Fit after recovery and above the noise floor. Check for beating, edge effects, gain drift and structured residuals. Record the fitted interval, τ, uncertainty and residuals. Estimate repeatability across shots. A direct damped-sinusoid fit is a useful cross-check for an isolated component.

Calibration and interpretation

Electrical calibration

Use a stable reference generator and a verified meter or oscilloscope. Check the trigger-to-sample offset and its repeatability with a timing marker. Check frequency against tones near the operating band and determine whether the sample clock needs a scale correction. Verify offset and gain at several known levels for each range used, then check amplitude response through the detector electronics where a test input is available. Keep factory calibration active for voltage-mode checks and document the conversion when storing raw counts. [5–7]

Frequency to mass calibration

For a fixed electrostatic orbit and energy per charge U, the ion speed is $\sqrt{(2qU/m)}$, where q is the ion charge and m is its mass. If the effective path length and harmonic order stay fixed, the observed frequency scales as $1/\sqrt{(m/z)}$, where z is the charge number. This gives a starting calibration with a single coefficient C. Use the same electrode and injection settings for standards and unknowns. [1–4, 11]

$$f \propto 1/\sqrt{(m/z)} \qquad m/z = C/f^2$$

Use a known ion to initialize C and multiple identified references spanning the working range to verify the model. For example, the sodium source in the September slides can supply a reference when the ion and charge state are confirmed. Compute C for each calibrant, check residuals and repeatability, and use an independent reference when available. Check whether each peak is a fundamental, a harmonic, or an electrical artifact before assigning m/z.

The sketch values 39 and 120 kHz give $C \approx 5.62 \times 10^{11}$ (Da/e)·Hz². This illustrates the calculation. Use measured reference frequencies and accepted ion masses for the real fit. Save coefficients, calibration conditions, validity range, reference residuals and uncertainty with the mass spectrum. An unassigned or out-of-range peak can still be reported in Hz.

Pressure calibration

For a fixed probe ion and a controlled gas at known pressure, an empirical first model is $\gamma = \gamma_0 + Kp$, where γ_0 is the baseline decay rate, p is pressure and K is the sensitivity for that ion–gas combination. Fit repeated measurements against an independent pressure reference and test the linear range. Geometry, ion dephasing and electronics can also affect the observed decay. [2–4]

For residual gas partial pressures inferred from spectral signals, measure species-specific response under known gas composition, source conditions, preamplifier gain and processing settings. Account for overlapping fragments and charge states. In a mixture, the decay of one ion can depend on several gas species. Report frequency, calibrated m/z and decay until the pressure response is established.

Bench validation and references

The checks below are proposed starting criteria. Set the final error limits with the lab using the required mass accuracy, pressure range and shot cadence.

1. **Verify the trigger with high voltage disabled** Scope the timing output and the conditioned edge. Compare source pulse count with accepted shots over 100 gated pulses and at the intended operating cadence. Measure delay, jitter and missed triggers. Confirm the script records a timeout when the pulse is withheld.
2. **Verify acquisition and replay** Inject a known tone near the expected band. Check trigger alignment, sample count, usable duration and clipping. Use synthetic single-tone, two-tone and damped traces to test the analysis. Initial targets for a clean synthetic case are frequency within one unpadded FFT bin and decay time within 10% of the input.
3. **Measure background and a known ion** Collect a pulse-only background with ion production disabled, then repeat with the sodium source or another confirmed calibrant. Compare raw traces and spectra, choose blanking and fit windows, and check that the assigned peak is repeatable and distinguishable from pickup. Report the achieved cadence and mass residuals.
4. **Package a reproducible run** Include the wiring diagram, script and settings, one raw and one background capture with metadata, analysis plots, and calibration evidence. Confirm the final run uses the agreed frequency band, post-trigger interval and repetition rate.

Technical references

[5] Red Pitaya, Original STEMlab 125-14 hardware and electrical specifications.

[6] **Red Pitaya,** external trigger example, acquisition command reference, and synchronized pulse example.

[7] **Red Pitaya,** Deep Memory Mode command reference and Deep Memory Acquisition example.

[8] Räcke et al. (2018), Detection of small bunches of ions using image charges.

[9] Tektronix, Floating measurements selection guide.

[10] NumPy real FFT frequency bins and SciPy Blackman–Harris window.

[11] Ishihara, Toyoda and Matsuo (2000), Perfect space and time focusing ion optics for multiturn time of flight mass spectrometers.